



GUJARAT TECHNOLOGICAL UNIVERSITY
L. D. COLLEGE OF ENGINEERING



Autonomous Green Hydrogen Plant Controller Using Fuzzy Logic

Subject: Fuzzy Logic and Genetic Programming | Code: BE05000461

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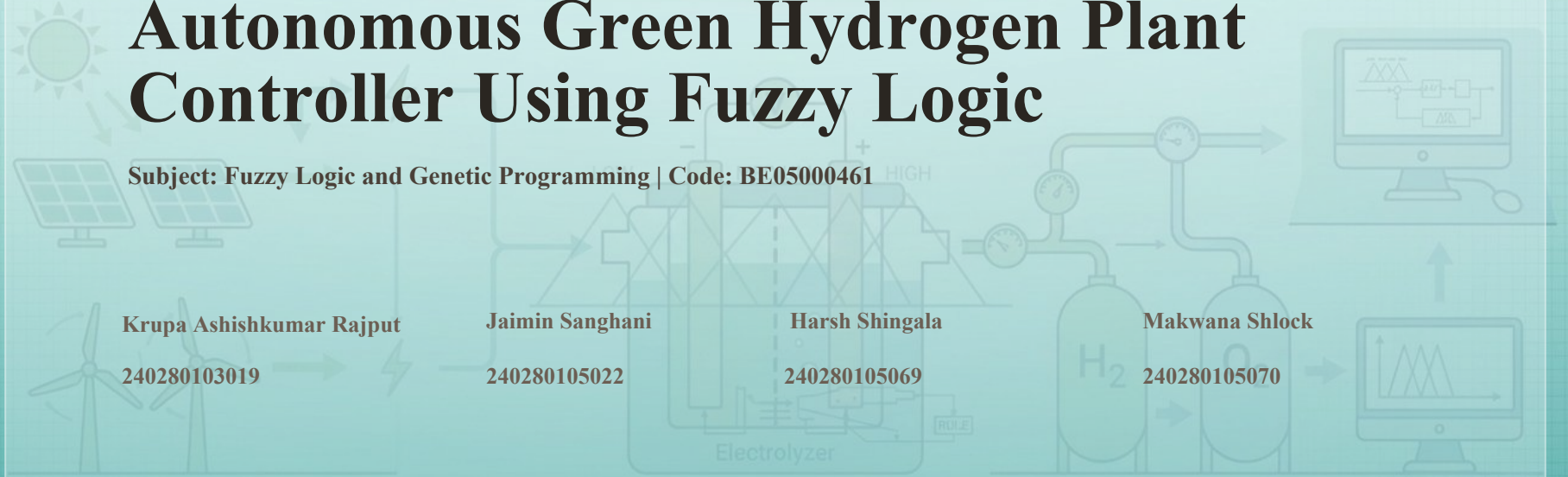
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The Imperative for Green Hydrogen

Green hydrogen is produced via electrolysis powered by renewables, offering a zero-carbon energy carrier for global decarbonization. Unlike grey or blue hydrogen, the green route leaves no fossil carbon behind — making it the only hydrogen vector fully aligned with a 1.5 °C pathway. This project builds the autonomous controller that such plants need to stay efficient under variable renewable input.

WHY WE ARE STUDYING IT



01

Eliminates fossil-fuel dependency in heavy industry.



02

Acts as long-duration, seasonal energy storage.



03

Enables clean heavy-duty transport and shipping.



Why a Fuzzy Controller — and What It Must Deliver

⚠ PROBLEM STATEMENT

P · 03

Three pain-points in today's green H₂ plants



Renewable Intermittency

Solar and wind fluctuations cause unstable electrolysis.



Non-linear Dynamics

Traditional PID controllers fail to manage complex chemical shifts.



Operational Inefficiency

Frequent manual intervention leads to energy losses.

✓ PROJECT OBJECTIVES

O · 03

What this project delivers



Design an Autonomous Controller using Fuzzy Logic

For real-time plant regulation.



Optimize Hydrogen Yield

By adapting to variable power inputs dynamically.



Ensure System Safety and Longevity

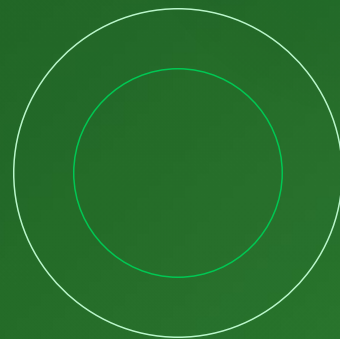
By minimizing electrolyzer stress.

CHAPTER

01 / 04

System Design

Sensors → Fuzzy Inference → Hydrogen Production Command

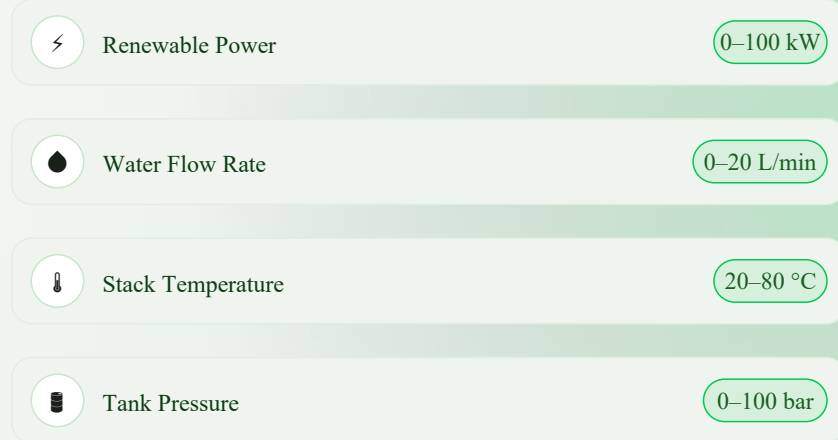


• 4 INPUTS · 1 OUTPUT · 11 RULES

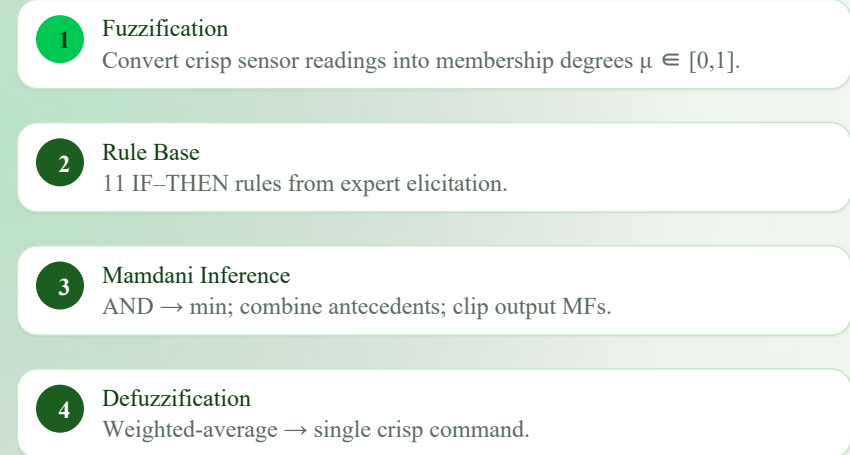
Proposed Fuzzy Logic Controller (FLC)

The Mamdani Fuzzy Logic Controller regulates Hydrogen Production Rate by monitoring real-time plant conditions across four input variables.

INPUT VARIABLES 4



PROCESSING PIPELINE 4 STAGES



Words Instead of Numbers

A linguistic variable uses words instead of only numerical values — letting the controller represent engineering knowledge in human-readable form.



02 / 04

Fuzzification

Triangular membership functions turn crisp sensors into degrees of truth $\mu \in [0,1]$.

➤ STEP 2 · INPUT 1 OF 4

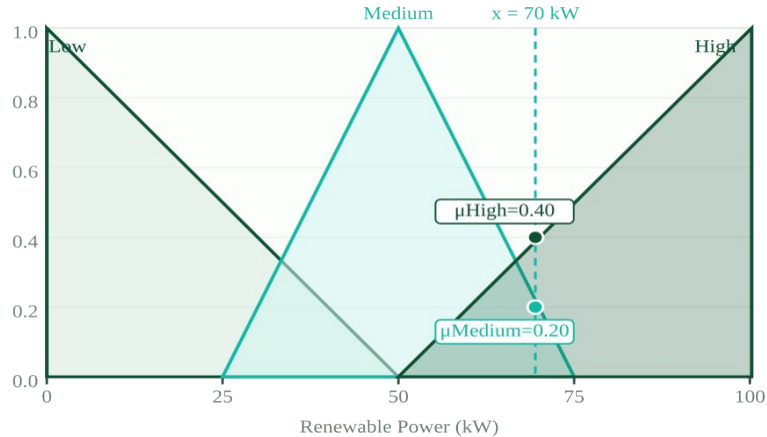
Membership Functions — Renewable Power

Triangular MFs over the universe 0–100 kW. Each term has parameters (a, b, c) defining its triangular shape.

TRIANGULAR MF PARAMETERS

Universe: 0–100 kW · 3 terms

3 TERMS



TRIANGULAR MEMBERSHIP FUNCTION

$$\mu(x) = \max(\min((x-a)/(b-a), (c-x)/(c-b)), 0)$$

TRIANGULAR MF PARAMETERS

Low a = 0, b = 0, c = 50

Medium a = 25, b = 50, c = 75

High a = 50, b = 100, c = 100

EXAMPLE

Renewable Power = 70 kW

$$\mu_{\text{Medium}}(70) = (75 - 70) / (75 - 50) \quad 0.20$$

$$\mu_{\text{High}}(70) = (70 - 50) / (100 - 50) \quad 0.40$$

$$\mu_{\text{Low}}(70) = 0 \text{ (outside Low region)} \quad 0.00$$

→ The reading is partially Medium (0.20) and partially High (0.40).

Membership Functions — Water Flow Rate

Triangular MFs over the universe 0–20 L/min — three terms: Low / Medium / High.

TRIANGULAR MF PARAMETERS

Low $a = 0, b = 0, c = 10$

Medium $a = 5, b = 10, c = 15$

High $a = 10, b = 20, c = 20$

EXAMPLE

Water Flow Rate = 14 L/min

$$\mu_{\text{Medium}}(14) = (15 - 14) / (15 - 10) \quad 0.20$$

$$\mu_{\text{High}}(14) = (14 - 10) / (20 - 10) \quad 0.40$$

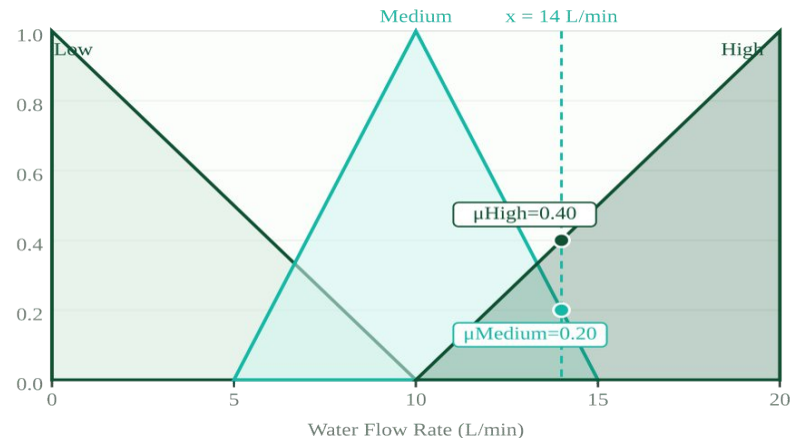
$$\mu_{\text{Low}}(14) = 0 \quad 0.00$$

→ Water flow is partially Medium (0.20) and partially High (0.40).

TRIANGULAR MF PARAMETERS

Universe: 0–20 L/min · 3 terms

3 TERMS



TRIANGULAR MEMBERSHIP FUNCTION

$$\mu(x) = \max(\min((x-a)/(b-a), (c-x)/(c-b)), 0)$$

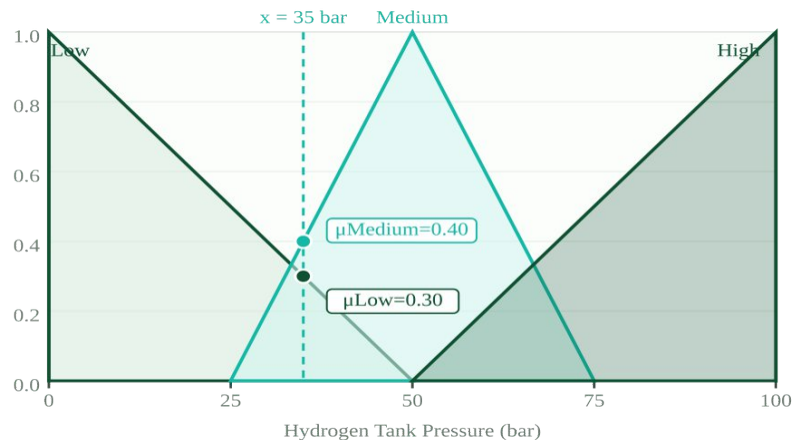
Membership Functions — Hydrogen Tank Pressure

Triangular MFs over 0–100 bar. Pressure is the safety-critical input — High pressure forces the controller to reduce output.

TRIANGULAR MF PARAMETERS

Universe: 0–100 bar · 3 terms

3 TERMS



TRIANGULAR MEMBERSHIP FUNCTION

$$\mu(x) = \max(\min((x-a)/(b-a), (c-x)/(c-b)), 0)$$

TRIANGULAR MF PARAMETERS

Low a = 0, b = 0, c = 50

Medium a = 25, b = 50, c = 75

High a = 50, b = 100, c = 100

EXAMPLE

Tank Pressure = 35 bar

$$\mu_{\text{Low}}(35) = (50 - 35) / 50 \quad 0.30$$

$$\mu_{\text{Medium}}(35) = (35 - 25) / (50 - 25) \quad 0.40$$

$$\mu_{\text{High}}(35) = 0 \quad 0.00$$

→ Tank pressure is partially Low (0.30) and partially Medium (0.40).

Membership Functions — Electrolyzer Stack Temperature

Triangular MFs over 20–80 °C. Three terms: Low / Normal / High — "Normal" is the safe operating band.

TRIANGULAR MF PARAMETERS

Low a = 20, b = 20, c = 40

Medium a = 30, b = 50, c = 70

High a = 60, b = 80, c = 80

EXAMPLE

Stack Temperature = 50 °C

$\mu_{\text{Normal}}(50) = 1.00 \leftarrow \text{peak}$ **1.00**

$\mu_{\text{Low}}(50) = 0$ **0.00**

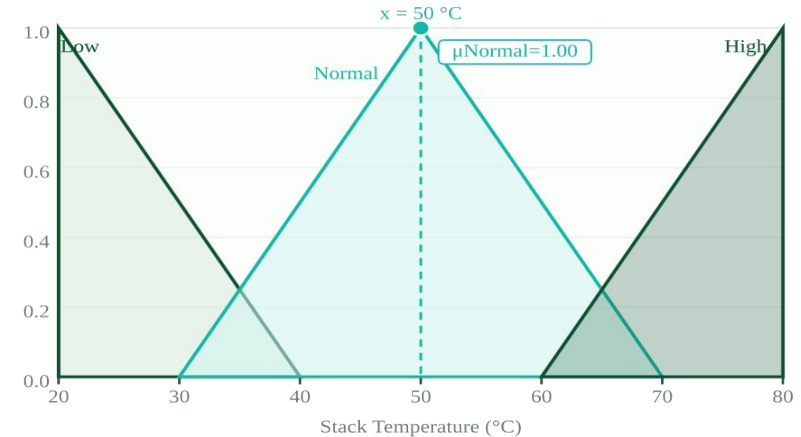
$\mu_{\text{High}}(50) = 0$ **0.00**

→ The electrolyzer stack is completely in the Normal temperature region.

TRIANGULAR MF PARAMETERS

3 TERMS

Universe: 20–80 °C · 3 terms



TRIANGULAR MEMBERSHIP FUNCTION

$$\mu(x) = \max(\min((x-a)/(b-a), (c-x)/(c-b)), 0)$$

🔧 STEP 2 · OUTPUT VARIABLE

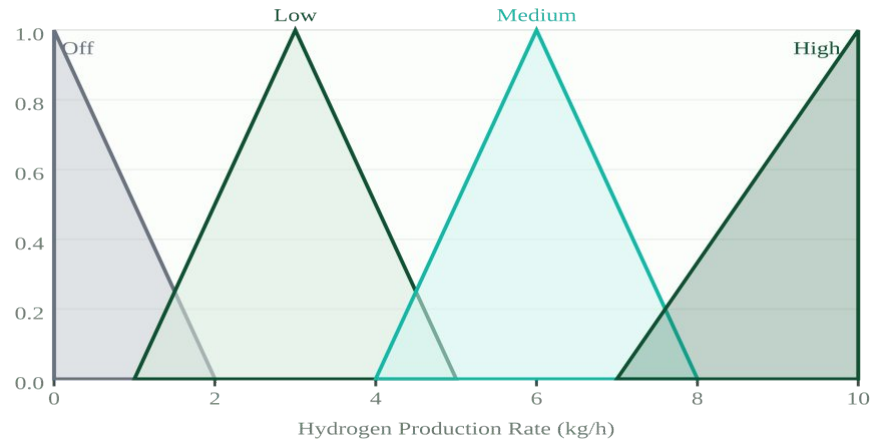
Output Variable — Hydrogen Production Rate

Triangular MFs over 0–10 kg/h — four terms: Off / Low / Medium / High. The controller's crisp command comes from defuzzifying this single output.

TRIANGULAR MF PARAMETERS · 4 TERMS

Universe: 0–10 kg/h · 4 terms

4 TERMS



PEAK VALUES FOR DEFUZZIFICATION

Peak values used as representative points for weighted-average defuzzification: Off→1, Low→3, Medium→6, High→10.

TRIANGULAR MF PARAMETERS

Off	$a = 0, b = 0, c = 2$	●
Low	$a = 1, b = 3, c = 5$	●
Medium	$a = 4, b = 6, c = 8$	●
High	$a = 7, b = 10, c = 10$	●

$$\mu(x) = \max(\min((x-a)/(b-a), (c-x)/(c-b)), 0)$$

◆ OUTPUT INTERPRETATION

- Off 0–2 kg/h — Production stopped
- Low ~3 kg/h — Low production
- Medium ~6 kg/h — Moderate production
- High ~10 kg/h — Full production

03 / 04

Knowledge Base & Inference

11 expert IF–THEN rules → Mamdani inference → firing strengths.

IF IF IF IF IF
IF IF IF IF IF
IF IF IF IF IF
IF IF IF IF IF

Fuzzy IF–THEN Rule Base — 11 Rules

Expert elicitation — five columns per rule (4 antecedents + 1 consequent). Color-coded by output term.

LEGEND

High

Medium

Low

Normal

Off

Rule	Renewable Power	Water Flow	Tank Pressure	Temperature	Output
R1	High	High	Low	Normal	High
R2	High	Low	Low	Normal	Medium
R3	Medium	Medium	Low	Normal	Medium
R4	Medium	Low	Low	Normal	Medium
R5	Low	High	Low	Normal	Medium
R6	Low	Medium	Low	Normal	Low
R7	Low	Low	Low	Normal	Low
R8	High	High	Medium	Normal	High
R9	High	Medium	Medium	Normal	Medium
R10	Low	Low	High	Normal	Off
R11	Medium	Any	Any	High	Low

Mamdani Inference — Evaluate → Combine → Clip

1 STEP 1 Evaluate Antecedents ♦

Read the membership degree of each input term.

For each input variable, find how strongly the crisp reading belongs to its Low, Medium, High (or Normal) linguistic term — i.e. $\mu \in [0,1]$.

2 STEP 2 Combine Using AND ♦

For AND: $w = \min(\mu_1, \mu_2, \mu_3, \mu_4)$.

The firing strength of a rule is the minimum of its antecedent memberships — the weakest antecedent gates the rule.

3 STEP 3 Apply Implication ⚙

$$\mu'_{\text{out}}(y) = \min(w, \mu_{\text{out}}(y))$$

The rule's firing strength clips the output membership function — values above w are flattened down to w .

WORKED EXAMPLE

Live Plant Reading

VARIABLE	READING	TERM
Renewable Power	70 kW →	$\mu_{\text{High}}(0.40)$
Water Flow Rate	14 L/min →	$\mu_{\text{High}}(0.40)$
Tank Pressure	35 bar →	$\mu_{\text{Low}}(0.30)$
Stack Temperature	50 °C →	$\mu_{\text{Normal}}(1.00)$

SAMPLED AT CONTROLLER TICK T_0

FIRING STRENGTH · RULE R1

$w_1 = \min(0.40, 0.40, 0.30, 1.00) = 0.30 \rightarrow$ clips the High output MF

04 / 04

Worked Example — End to End

From a live sensor reading → 8.67 kg/h autonomous command.

PIPELINE

Fuzzify → Evaluate Rules → Aggregate → Defuzzify

♥ STEP 1 — FUZZIFY

Fuzzify a Live Plant Reading

♥ SENSOR SNAPSHOT — LIVE PLANT READING

RENEWABLE POWER
70 kW

WATER FLOW RATE
14 L/min

STACK TEMPERATURE
50 °C

HYDROGEN TANK PRESSURE
35 bar

Table A · Renewable Power= 70 KW

Universe 0–100 kW

Term	Calculation	μ
Low	outside Low region	0
Medium	$(75 - 70) / (75 - 50)$	0.20
High	$(70 - 50) / (100 - 50)$	0.40

Table B · Water Flow Rate=14 L/min

Universe 0–20 L/min

Term	Calculation	μ
Low	outside Low region	0
Medium	$(15 - 14) / (15 - 10)$	0.20
High	$(14 - 10) / (20 - 10)$	0.40

Table C · Tank Pressure=35 bar

Universe 0–100 bar

Term	Calculation	μ
Low	$(50 - 35) / 50$	0.30
Medium	$(35 - 25) / (50 - 25)$	0.40
High	outside High region	0

Table D · Stack Temperature=50 °C

Universe 20–80 °C

Term	Calculation	μ
Low	outside Low region	0
Normal	peak	1.00
High	outside High region	0

STEP 3 — RULE EVALUATION

Evaluate the Rules That Fire

Only rules whose antecedents all have non-zero membership contribute — Mamdani AND $\rightarrow$ min.

NON-ZERO MEMBERSHIPS FROM STEP 1

RENEWABLE POWER
Medium = $0.20 \cdot \text{High} = 0.40$

WATER FLOW RATE
Medium = $0.20 \cdot \text{High} = 0.40$

TANK PRESSURE
Low = $0.30 \cdot \text{Medium} = 0.40$

STACK TEMPERATURE
Normal = 1.00

◆ Fired Rules — 0 of 11 rules fire

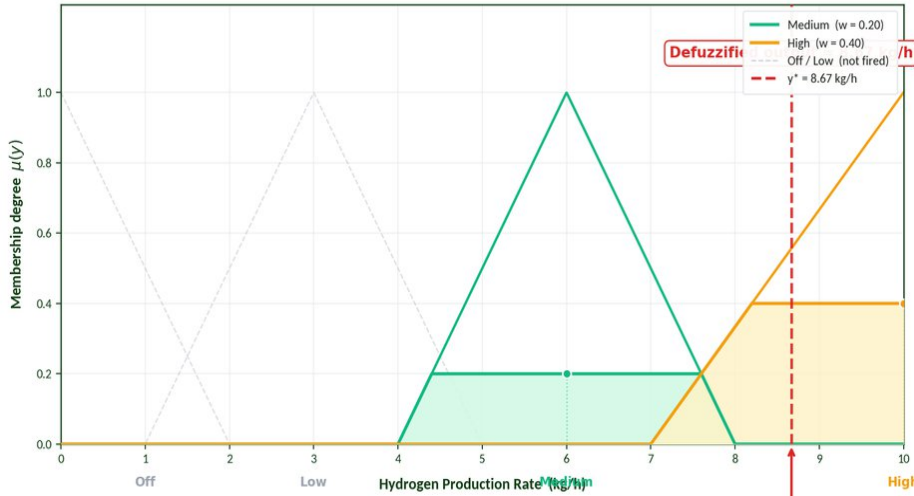
AND = min · Mamdani inference

Rule	Antecedent Calculation	Firing Strength	Output
R1	$\min(0.40, 0.40, 0.30, 1.00)$	0.30	High
R3	$\min(0.20, 0.20, 0.30, 1.00)$	0.20	Medium
R8	$\min(0.40, 0.40, 0.40, 1.00)$	0.40	High
R9	$\min(0.40, 0.20, 0.40, 1.00)$	0.20	Medium

Note: Other rules — all remaining rules have firing strength = 0 and therefore do not contribute to the output.

◆ STEPS 5–6 — AGGREGATION & DEFUZZIFICATION

Aggregate the Outputs & Defuzzify



Defuzzification

WEIGHTED AVERAGE:

Output Term	w	Peak (kg/h)	w × Peak
Medium	0.20	6	1.20
High	0.40	10	4.00

$$\Sigma w = 0.20 + 0.40 = 0.60$$

$$\Sigma(w \times \text{peak}) = 1.20 + 4.00 = 5.20$$

$$y^* = \Sigma(w_i \times \text{peak}_i) / \Sigma w_i = 5.20 / 0.60$$

$$y^* = 8.67 \text{ kg/h}$$

For Renewable Power = 70 kW · Water Flow = 14 L/min · Tank Pressure = 35 bar · Stack Temp = 50 °C, the controller recommends producing hydrogen at approximately **8.67 kg/h**.

✦ CRISP OUTPUT — HYDROGEN PRODUCTION RATE :-

$$y^* = 8.67 \text{ kg/h}$$

Key Takeaways & Further Reading

Key Takeaways

6 stages · 1 crisp output

1 ■

Crisp → Fuzzy
Real plant variables rarely have sharp boundaries; MFs represent degrees of truth.

2 ✿

Fuzzification
Crisp sensor readings become membership degrees $\mu \in [0,1]$.

3 ■

Rule Base
Expert IF–THEN rules connect input conditions to production decisions.

4 ◆

Mamdani Inference
MIN combines AND antecedents and determines rule firing strength.

5 ⚙

Aggregation
MAX combines the outputs of fired rules.

6 ⚙

Defuzzification
Weighted average converts the fuzzy result into a single crisp command.

FINAL SAMPLE RESULT

Renewable Power **70 kW** · Water Flow **14 L/min** · Tank Pressure **35 bar** · Stack Temp **50 °C** → $y^* = 8.67 \text{ kg/h}$

[7] BE05000461 — Fuzzy Logic and Genetic Programming, PBL Details.

COURSE

References

7 sources ·
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Course

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- [5] MathWorks, *MATLAB Fuzzy Logic Toolbox User's Guide*. TOOLING
- [6] G. J. Klir & B. Yuan, *Fuzzy Sets and Fuzzy Logic: Theory and Applications*, Prentice Hall. TEXTBOOK